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Section: 4C

Table of contents

OBJECTIVE

The objective of this project was to design and implement a scalable, three-layer hierarchical enterprise network using Cisco Packet Tracer. The project aimed to achieve automated IP assignment across different subnets via DHCP relay, robust dynamic connectivity using RIP (Routing Information Protocol), and logical zone segmentation for optimized traffic management.

NETWORK ZONES OVERVIEW

1. Access Zone (Pink): The Access Zone serves as the entry point for end-user devices. It provides connectivity for PCs and laptops, utilizing DHCP relay to request IP addresses from the central server.

• DHCP Relay: Configured using the ip helper-address command on the access router to forward broadcast requests to the server in the Distribution Zone.

• Connectivity: Devices are segmented into their own subnet to maintain efficient broadcast domains.

2. Core/Backbone Zone (Blue): This zone acts as the transit backbone of the network, interconnecting the Access and Distribution layers.

• Dynamic Routing: Implemented RIP to ensure automatic convergence and path redundancy between zones.

• Redundancy: Dual-path connections were established to ensure that traffic is automatically rerouted if a primary link fails.

3. Distribution Zone (Green): This zone houses the centralized network services, including the primary DHCP server.

• DHCP Services: Hosts a multi-pool DHCP server capable of assigning unique IP ranges to different zones based on relay agent requests.

• Connectivity: Connects to the Core layer to provide the 'return path' for all routed traffic.

TECHNOLOGIES USED

• Cisco Packet Tracer: Utilized for physical design, logical addressing, and simulation of packet flow.

CISCO USAGE

Cisco Packet Tracer provided the environment for building the three-layer hierarchical model. Key applications included:

• Network Design: Logical topology creation using standard hierarchical Cisco models.

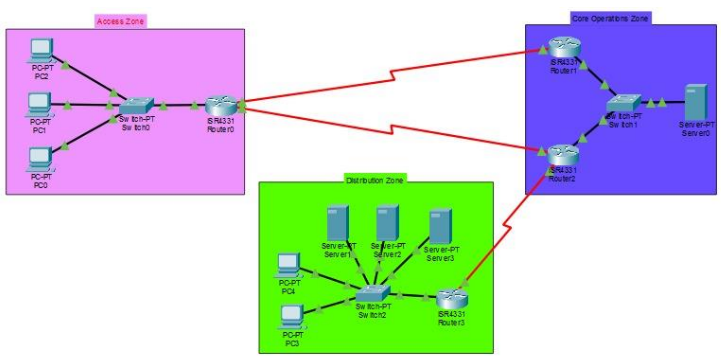
• Dynamic Routing (RIP): Configured routers to exchange routing tables automatically, verified via show ip route to confirm 'R' entries.

• DHCP Relay: Configured ip helper-address to bridge the gap between DHCP broadcast domains and the server.

• End-to-End Testing: Used Simulation Mode with ICMP filters to visualize packet transit across backbone routers.

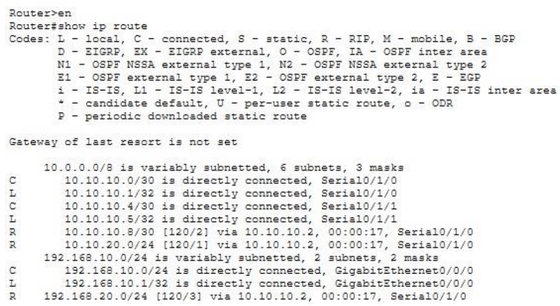
TOPOLOGY

Figure 1: Enterprise Network Topology illustrating the three-layer hierarchical segmentation of the Access Zone (Pink), Distribution Zone (Green), and Core Operations Zone (Blue).



CONFIGURATION, RESULTS & TESTING

• **RIP:** Verified connectivity through show ip route on all routers; successful 'R' entries confirmed dynamic path learning.



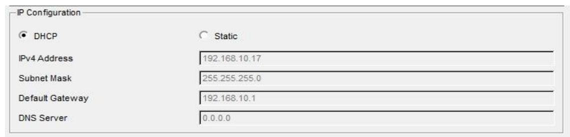
```
Router>en
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
C       10.10.10.0/30 is directly connected, Serial0/1/0
L       10.10.10.1/32 is directly connected, Serial0/1/0
C       10.10.10.4/30 is directly connected, Serial0/1/1
L       10.10.10.5/32 is directly connected, Serial0/1/1
R       10.10.10.8/30 [120/2] via 10.10.10.2, 00:00:17, Serial0/1/0
R       10.10.20.0/24 [120/1] via 10.10.10.2, 00:00:17, Serial0/1/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
R    192.168.20.0/24 [120/3] via 10.10.10.2, 00:00:17, Serial0/1/0
```

• **DHCP:** Successfully implemented across zones; PC configuration confirms successful assignment of IPs from the specific pools defined on the Distribution Zone server.

Figure 2: DHCP Address Assignment Confirmation on Access Zone Device.



• **Verification:** Successful ping tests from the Access Zone (Pink) to the Distribution Zone (Green) confirmed full network convergence.

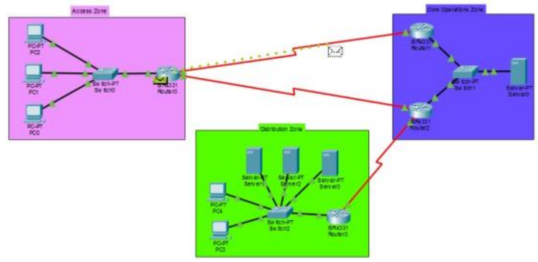
Major Commands Used:

• **RIP:** router rip, network <address>

• **DHCP Relay:** ip helper-address <server-ip>

• **Verification:** show ip route, show ip interface brief

Figure 3: Simulation mode showing active ICMP packet transit from the Access Zone to the Distribution Zone, validating end-to-end network connectivity.



CHALLENGES & LEARNING

Challenges Faced:

• DHCP Relay Configuration: Initial challenges in ensuring the ip helper-address was applied to the correct router interface facing the end-users.

• Routing Table Convergence: Ensuring RIP propagated routes for all subnets across the Core backbone.

• IP Conflicts: Troubleshooting 'address already used' errors by ensuring unique static IPs for servers and routers.

Lessons Learned:

• Hierarchical Logic: The importance of separating Access, Distribution, and Core layers for network scalability.

• Troubleshooting Methods: Mastered the use of simulation mode to isolate traffic and show ip route to verify pathing.

• Relay Mechanics: Understanding that routers block broadcasts by default, necessitating the helper-address relay agent.

CONCLUSION

This project successfully demonstrated the design and implementation of a professional, three-layer hierarchical network. By incorporating dynamic routing via RIP and centralized DHCP relay, the network achieved reliable automated IP allocation and robust connectivity.

Outcomes:

• A fully operational, redundant network topology.

• Practical mastery of Cisco routing protocols and DHCP relay mechanisms.

FUTURE RECOMMENDATIONS

To evolve this network from a functional base configuration into a high-availability, enterprise-grade production environment, the following implementations are recommended for future iterations:

1. Secure Connectivity (VPN): Site-to-Site VPN Deployment: To further secure communication across the untrusted backbone, future iterations of this network will include IPsec site-to-site VPN tunnels between remote routers.

2. Mobility & Wireless Infrastructure: Wireless Integration: To support a mobile workforce, the Access Zone infrastructure can be extended by deploying Lightweight Access Points (LAPs) managed by a central Wireless LAN Controller (WLC).

3. Management & Resiliency: Network Monitoring: Implementing centralized syslog servers and NTP (Network Time Protocol) would improve operational visibility and correlate security events with precise timestamps.